

On-Chain Reputation Ranking by Adaptive Weighted PageRank

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Abstract—Reputation is a fundamental concept in human social interactions and inter-relationship. Blockchain and Web3 have been rapidly growing, involving multi-millions of users, billions of transactions, and hundreds of millions of addresses onchain (i.e. on-blockchain). Decentralized identity and onchain reputation will play a critical role in defining each individual’s persona, and natively supporting onchain entities to build trust in the Web3 space, while preserving user privacy and autonomy. This paper presents a novel reputation ranking method to address the problem. We leverage adaptive weighted PageRank algorithms to assess and rank the reputation of participants within various public blockchains. We assign reputation scores to onchain entities based on their historical transaction behaviors, taking transaction volume, activeness, and network contributions into account. In addition to describing the theoretical framework of our reputation ranking, this paper provides empirical insights into its implementation on transactional datasets of Ethereum and BNB Chain. The results and findings presented herein offer a foundation for further research and development of onchain reputation systems in public blockchains and its applications in decentralized finance and Web3 ecosystems.

Index Terms—adaptive Pagerank, Blockchain, onchain reputation, PageRank, reputation ranking, Web3

I. MOTIVATION OF REPUTATION RANKING IN WEB3

Since the paper of L. Page et. al. [1], PageRank and similar ranking algorithms have been extensively applied for web search engines (by Google and others), American school-football conference ranking, social network analysis (e.g. citation and collaboration network, etc), Twitter’s recommendation and other use cases. If we view a blockchain ecosystem as a social network made of onchain accounts and their transactions, then we can apply Pagerank and mathematical ranking algorithms to compute reputation scores of onchain entities, extracting the topology of the transaction graph, comprehensive social insights and inter-relationship within the blockchain ecosystem.

Reputation is a fundamental concept that has been central to human social interactions and inter-relationship. It refers to the collective, transactions, beliefs or opinions held by others about a person or entity, and it can have a significant impact on trust, credibility, and success, especially in the digital world. With the rapid growth of the Web3 space, involving multi-millions of users, billions of transactions, and hundreds of millions of addresses on various blockchains,

by a decentralized manner, measuring the specific reputation of a Web3 entity has become more and more important. Coinbase is calling onchain reputation builders for its Base Ecosystem Fund. The Base’s request for builders claims that “decentralized identity and reputation will play a critical role in defining each individual’s onchain persona, and reputation protocols (e.g. Google Pagerank type) should natively support onchain entities to build trust onchain, while preserving user privacy and autonomy.”

This study develops a novel pairwise ranking methodology for comprehending and quantifying authentic social interactions, reputation ranking of accounts within public blockchain ecosystems. We develop an efficient data pipeline and computation process tailored for the analysis of big datasets from public blockchains, providing a valuable resource for future research endeavors. The outcomes and insights presented in this paper serve as a cornerstone for advancing research and development in the realm of reputation ranking and scoring systems for public blockchain networks and their practical applications in decentralized finance and Web3.

II. RELATED WORKS

In 2018, NEM (XEM) was the first blockchain, established a theoretical framework of ranking from blockchain-based transactions and its application for blockchain consensus mechanism. In the technical paper [4], the NEM developer introduced a novel consensus mechanism based on reputation system of validator nodes and proof of importance metrics measured by NCDawareRank ranking algorithm [3].

In 2019, Thuat Do et. al. [6] introduced delegated proof of reputation, a variation of Proof of Stake consensus based on onchain reputation ranking. The authors analyzed Pagerank, HodgeRank and applied to calculated reputation scores. Later, Dung Do and Thuat Do [7] continued investigation on application of Pagerank and HodgeRank for social credit measurement from onchain transactions.

Colendi introduced a mathematical ranking-based algorithm to calculate credit score, then applied for microlending industry. The firm acquired telco data, social media and trusted partners, then scoring user credit ranking, offering Colendi Score it on its platform connecting lenders and borrowers. The algorithm process is as follows:

- 1) Generate a graph-typed network of users with two directed edges:
 - a) Social connectivity
 - b) Activities
- 2) Calculate the Colendi Score of each person
- 3) Normalize all scores on the entire network.

The protocol investigates W_{ij} , the “social connectivity strength” between two nodes i and j to define social relation strength and contribution to social credibility. It is a compound of relation scalar strength between two nodes, denoted by $W_{ij}^{relation}$ and dependent bilateral social relation activity, denoted by $W_{ij}^{activity}$. It reads (for a weighting factor β)

$$W_{ij} = \beta * W_{ij}^{activity} + (1 - \beta) * W_{ij}^{relation}.$$

The Colendi Score of a node i is computed as

$$S_i = \alpha \left(\frac{1 - d}{N} + \frac{d * \sum_j \frac{S_j * W_{ij}}{S_{max}}}{\sum_j W_{ij}} \right)$$

where d the damping factor, N the total number of users (or nodes), α the normalizing coefficient, S_j^i the Colendi Score of the j -th neighbor of the node i , S_{max} the maximum Colendi Score of all nodes.

III. REPUTATION RANKING METHODOLOGY

A. The Adapted PageRank Algorithm (APA)

PageRank, initially developed by Larry Page and Sergey Brin [1], is a fundamental algorithm in the field of information retrieval and web search. It revolutionized the way web pages are ranked and is the basis for many modern search engines, including Google. PageRank assigns numerical values to web pages based on the number and quality of links pointing to them. In essence, it measures a webpage’s authority and importance in the vast network of interconnected web pages, making it a crucial tool for organizing and retrieving relevant information from the web. This algorithm’s significance extends beyond search engine, as it has been adapted and extended to various applications, including the development of reputation systems, where it can be used to assess the credibility and influence of individuals or entities within social networks or graphs.

The adapted PageRank algorithm, firstly proposed by Agryzkov et. al. [9], offers a model for establishing node rankings within an urban network, considering the available data. While originally designed for urban networks, this centrality measure can be applied more broadly to spatial networks or social networks with data. Building upon the fundamental concept of the PageRank vector, this algorithm incorporates the significance of data from various sources, then modifies the construction of the matrix used to determine node rankings.

In the APA model, the data matrix is constructed in a reminiscent manner of the original PageRank vector concept. In the model, a random walker has the ability to transition between connected nodes by following the local links provided by the

network. Additionally, the walker can jump between nodes that are not directly connected, with the same probability, irrespective of their topological distance, which is defined as the number of nodes in the walk.

In the implementation of the APA centrality, let $G = (N, E)$ be a graph, where $N = \{1, 2, \dots, n\}$, $0 < n \in \mathbb{Z}$, and $E = \{(i, j)\}$ the set of all edges (links) connecting nodes i and j . The adjacency matrix of G is $A_{n \times n} = (a_{ij})$, where:

$$a_{ij} = \begin{cases} 1 & \text{if } (i, j) \in E, \\ 0 & \text{otherwise} \end{cases}$$

The transition matrix P is constructed by $P = (p_{ij})$, where $p_{ij} = \frac{1}{\sum_{i=1}^n a_{ij}}$, representing, by columns, the probability of navigating from one page to another. The transition matrix exhibits important characteristics: non-negative, stochastic by columns, irreducible and primitive.

The central aspect of the model involves creating the data matrix D of size $n \times k$, where its n rows correspond to the n nodes within the network, and each of its k columns represents the attributes of the data objects under investigation. However, not all data characteristics may hold equal relevance to or influence on the specific question under analysis. Consequently, a vector w_0 is constructed in which the elements occupying row i corresponds to the multiplicative factor associated with property or characteristic k_i . This vector w_0 introduces a weighting mechanism for the data, allowing for the consideration of the entire dataset or subset of it. Then, we obtain $w = D \times w_0$, assigning a value to each node which represents the amount of data allocated to the node. Consequently, each node is associated with two distinct values: its degree, which pertains to the topology, and the data value assigned to it. For a more comprehensive explanation of how data is linked to the nodes, please refer to [9], [10].

We denote w^* as the normalized vector of w , and define

$$M = (1 - d)P + dV \quad (1)$$

The matrix $V \in \mathbb{R}^{n \times n}$ is characterized by having all of its components in the i -th row equal to w^* . The damping factor d is a fixed parameter associated with the concept of teleportation, commonly $d = 0.85$. The primary beauty of this algorithm lies in its construction of the data D and the weighted vector w_0 . The matrix D serves to numerically represent the dataset, while vector V determines the significance of each the factor or characteristics measured through D .

B. Adapting the PageRank to Blockchains

A blockchain-based transaction network can be viewed through the lens of graph theory, where nodes represent on-chain accounts (or interchangeably, addresses), and edges represent transactions between them. Our objective is to employ the APA model within the context of the public blockchain transaction network to create an effective reputation scoring system. Do Huu-Dung and Thuat Do [7] proposed a solution utilizing mathematical pairwise ranking methods like PageRank and HodgeRank to rank addresses on the Ethereum

blockchain. The method yielded promising ranking results, particularly when identifying top accounts and exhibiting a high correlation with in-degree, in-transfer indicators, which reflected the quality of connections. It's important to note that in the design, the authors made the assumption that all edges carried equal values. Users did not display preferences for interacting with specific addresses. This assumption is not entirely suitable in Web3 (or the Internet of Blockchain). Node reputation varies based on many factors such as activity levels, transferred value, participating communities, and others. This serves as the primary motivation behind our proposal to reconfigure matrices V and P as detailed in Eq. 1, with the aim of assigning greater weights to nodes that play more pivotal roles within the transaction network.

Daily on public blockchain networks, multi-millions of transactions serve various purposes: value transfers up to million dollar, contract internal calls, social interactions with no transferred value. It's evident that the larger the transaction value, the more it signifies trust among transaction participants. Addresses receiving numerous highly-valued transactions tend to gain higher trust within the network. Moreover, due to rapid technology advancements and the rise of decentralized applications, nodes with recent significant activities are assumed to have greater importance within the network. For instance, on any Web3 ecosystem, DeFi applications play the most crucial role in user acquisition, engagement and ecosystem growth. Thus, DeFi communities and DeFi contracts should receive higher evaluation. We must consider a time-based factor to enhance the activeness attributes of network components. Based on these characteristics, we propose the incorporation of time-aware and value-aware features when constructing data matrices. In essence, this involves assigning weights to nodes and edges within the network.

1) *Reputation ranking intuitions*: The reputation score of a public address is the probability of a random unbiased account interacting with that address at any given moment in an infinite random interaction, or simply, measures the importance of the account within the considered Blockchain ecosystem. Some intuitions for reputation ranking are:

- More in-transfers or in-degree (e.g. receiving value), then higher reputation score;
- In-transfers from a highly reputable account result in greater reputation score than receiving from a low reputable account;
- Out-transfers or out-degree (e.g. sending value) result in a lower reputation score;
- Weighting transactions: greater valued transactions gain higher weights, and more recent transactions are more meaningful than the old ones.

2) *Time-based weighting mechanism*: Onchain transactions are securely recorded in a distributed ledger with unchangeable timestamps. Leveraging transaction timestamps, we can collect information about an account's activities in various periods of time, the diversity of its interactions within the network. We propose a time-based weighting mechanism to assess the

activeness engagement level of an account in a blockchain ecosystem as follows:

$$X_i = \sum_{j=1, i \neq j}^N a_{ij} (\max_{t \in H_{ij}} T(t)), \quad (2)$$

where the timestamp transformation $T(t) = \frac{1}{1 + e^{-a(t-t_0)}}$ is a variation of the logistic function known for its distinctive S-shaped curve, H_{ij} is the set of all timestamped transactions from node i to node j . Time-based weights start slowly, then rise significantly in the middle, and ultimately plateau as they approach the upper limit. Consequently, transactions occurred in the past imply lower activeness for the involved accounts, hence receive lower weights. Conversely, recent transactions are considered more significant and receive greater weights. The output values of the transformation function fall within the range of 0 to 1, reducing computational complexity and making it suitable for statistical probability models. The growth-rate and midpoint parameter can be adjusted to provide flexible transformations that align with different input data.

3) *Value-based weighting mechanism*: We assume that transactions with greater values carry greater importance within the network. Consequently, transactions with higher values will be assigned greater weights. To evaluate the level of importance of transactions, we introduce a value transformation function as below:

$$Y_{ij} = \sum_{t \in H_{ij}, z \in Q_{ij}} T(t)V(z), \quad V(z) = \frac{2}{1 + e^{-bz}} - 1, \quad (3)$$

where Q_{ij} is the set of all valued transfers from node i to node j . The tanh function $V(z)$ is utilized to transform the transferred value of a transaction. Unlike the logistic function, which is often adjusted to be centered at zero by shifting, the tanh function inherently centers itself at zero. This attribute proves valuable in cases where we need to preserve the sign of the input data. Additionally, restricting weight values within the range of 0 to 1 serves to simplify computational tasks. As transaction weights are also affected by time, greater-valued and recently executed transactions will receive higher weights. To achieve this outcome, we implement the activeness factor $T(t)$ into the value weighting mechanism.

4) *Adapted PageRank algorithm for onchain reputation ranking*: Now, we are going to define our **adapted PageRank algorithm** for onchain reputation ranking and scoring problem. Let $PR(i)$ be the PageRank score of the node i . It is iteratively updated using the formula below:

$$PR(i) = (1 - d) \frac{X_i}{\sum_{i=1}^N X_i} + d \sum_{j \in \text{In}(i)} \frac{PR(j)Y_{ji}}{\sum_{k \in \text{Out}(i)} Y_{jk}} \quad (4)$$

where X, Y are the weighting functions defined in Equations 2 and 3, $\text{In}(i)$ is the set of nodes in-linked to node i , and $\text{Out}(i)$ is the set of nodes out-linked from i . The Perron-Frobenius theorem, by Pillai et. al. [11], holds a significance in the reputation ranking problem, ensuring the existence of an eigenvector associated with the dominant eigenvalue $\lambda = 1$,

given all entries are positive. This property allows an ordering or classification for these elements. In our case, given the construction of matrices P and V in Eq. 1, it is clearly a column-stochastic matrix, which guarantees the conditions to apply the Perron-Frobenius theorem.

IV. ONCHAIN REPUTATION RANKING: EXPERIMENTS AND RESULTS ON PUBLIC BLOCKCHAINS

A. Data Processing

In the context of our research paper, we implement a reputation scoring model on the Ethereum network, utilizing transaction data queried from [Ethereum datasets on Google Bigquery](#). Utilizing Spark within the data processing pipeline enables the harnessing of its distributed computing capabilities, scalability and user-friendliness to efficiently analyze and derive valuable insights from the continually expanding volumes of blockchain data [12]. In our case of Ethereum dataset, once raw data is obtained, the data processing phase involves cleaning and validating the collected data to eliminate duplicates, errors, and irrelevant information. Subsequently, the blockchain data is transformed into a structured format, such as DataFrames or RDDs (Resilient Distributed Datasets) suitable for Spark’s processing capabilities. Finally, the processed blockchain data is loaded into the memory of the computing cluster for the computation of reputation scores.

Because the data volume used in the proposed reputation model can be extensive, sometimes reaching hundreds of gigabytes, computing reputation scores requires robust parallel processing capabilities. GPUs or Spark clusters can be chose for these computations, each method having its own distinct advantages and drawbacks. GPUs are designed for highly parallel processing, boosting thousands of small cores optimized for matrix operations and data-parallel tasks. They excel in tasks demanding intensive numerical computations, allowing them to concurrently handle large-scale matrix operations and PageRank calculations. This parallelism proves particularly advantages for iterative algorithms like PageRank. Spark clusters, conversely, consist of multiple commercial CPUs, each equipped with multiple cores. They achieve parallelism through distributed computing but rely on CPUs which are versatile and capable of handling a wide array of tasks, they might not match GPUs in terms of performance for highly parallel computations. However, setting up GPUs clusters can require specialized expertise and be costly. In our experiment, we will implement both of these computation methods and conduct performance comparisons.

B. Reputation ranking results

1) *Computing Performance Comparison:* Table I illustrates the efficiency of GPU-based parallel computing when compared to CPU-based processing. These experiments were conducted using a serve with 4 Tesla T4 GPUs and the Intel Xeon processor 32 cores. The results indicate that for small datasets, GPU cluster processing speed is on par with CPU processing. However, as the graph size increases, CPU processing time

Data size (transactions)	CPU performance	GPU performance
10K	0.2s	0.35s
100K	0.27s	0.32s
1M	12.5s	0.43s
10M	70.5s	0.72s
100M	401.7s	0.69s

TABLE I

REPUTATION RANKING: PROCESSING PERFORMANCE COMPARISON

Pair of ranking methods	Scores
Simple PR ($d=0.85$) vs In-Value	0.385
Simple PR ($d=0.85$) vs In-Degree	0.522
Our APA ($d=0.85$) vs In-Value	0.432
Our APA ($d=0.85$) vs In-Degree	0.531
Our APA ($d=0.5$) vs In-Value	0.445
Our APA ($d=0.5$) vs In-Degree	0.567
Our APA ($d=0.5$) vs Simple PR	0.385
Our APA ($d=0.85$) vs Simple PR	0.381

TABLE II

KENDALL-TAU CORRELATION SCORES

risers quickly, while GPU processing performance remains relatively stable.

The utilization of GPUs or distributed computing introduces overhead due to data transfer between the CPU and GPU (or among distributed nodes). This overhead can offset potential benefits when dealing with small datasets. For larger datasets, parallel computing significantly enhances performance. In scenarios such as computing reputation scores on blockchain transaction networks, employing parallel computation can meet real-time computational demands.

2) *Validating reputation ranking:* Kendall’s Tau Correlation (KTC) is a statistical metric used to measure the difference between two ranking or ordering methods of the same list. KTC score is valuable for various fields and doesn’t assume data distribution, ranging between -1 and 1, with positive values indicating positive correlation and negative values indicating negative correlation, and 0 indicating no correlation. To validate the fit of our reputation ranking algorithms to the addressed ranking intuitions, we employ the Kendall-Tau method to measure the correlation between our proposed adapted Pagerank - APA model’s outcomes and some network’s statistics like degree, in-degree, in-value. Additionally, we perform a simple PageRank model (without weighting) on the network to gauge the effectiveness of both PageRank models. The correlation results are presented in Table II.

Table II demonstrates a significantly positive correlation scores between our algorithms and the in-degree & in-value statistics, showing a good fit to our intuitions, reliably reflecting importance of onchain accounts. These correlation is notably more effective than the simple PageRank algorithm when the weight mechanism is not applied. However, our experiments also reveal that the correlation scores should not be too high (being close to 1 makes reputation ranking similarly with normal statistical ranking methods), or too low (below 0.2 implies reputation ranking not fit the intuitions and not well-reflecting the insights from blockchain-based transaction graph). Furthermore, the ranking results indicate

that by employing the damping factor $d = 0.5$, we achieve better correlation scores than those of $d = 0.85$.

Figure 1 displays the histogram distribution of reputation scores based on our proposed adapted weighted PageRank and the simple PageRank. The results reveal that the weighted PageRank yields a significantly broader range of scores. Our experiments indicate that as the dataset size increases, the score range of the adaptive weighted PageRank widens considerably in comparison to the simple PageRank. This expansion allows for a more comprehensive ranking of onchain entities, particularly those with very low reputation scores. Notably, these low-scoring entities often constitute a substantial portion of the distribution.

V. ONCHAIN REPUTATION RANKING: APPLICATIONS, CONCLUSION AND FUTURE STUDIES

A. Applications of onchain reputation ranking

The reputation ranking algorithm is implemented on **Octan Network**, building a novel reputation ranking system and data analytics platform to establish trust and credibility in Web3 space, natively supporting onchain entities. In summary, onchain reputation ranking is potential to be applied in various scenarios throughout the Web3 space as follows.

- *Extracting social insights and inter-relationship* of on-chain entities within Web3 ecosystems.
- *Capturing user persona* without compromising privacy, offering a universal metric to qualify and segment users for targeted marketing and acquisition purposes
- *Investment research*: Reputation scores gives researchers a comprehensive and quantitative method to evaluate dapps, Web3 projects and their users for investment.
- *Onchain analytics*, understanding market dynamics and classification, user persona, ecosystem advancement, and growing trends.
- *Community governance*: Onchain reputation empowers Web3 projects to create a voting mechanism combining reputation score and staking amount, then build a more robust DAO, a better democratic decision-making & governance than current pure-staking models. Octan Network published an article to discuss “*How onchain reputation can enhance DAOs and prevent voting attacks*”, pointing to the **Tonado Cash’s governance attack** on May 2023.
- *Under-collateralized crypto-lending*: By establishing a reliable trustworthiness metrics in Web3, onchain reputation scores facilitate trust and credibility assessment, hence unleashing under-collateralized crypto-lending (now being impossible). This can be a big advancement, unleashing DeFi applications and mass adoption across blockchains.

B. Conclusion and future studies

Figure 2 presents a visualization of prominent dapps within the Ethereum network, with nodes sized by the cumulative reputation scores of their users, showing a good fit with common awareness on the dapps. It also indicates a reliable, quantitative evaluation for dapp quality and importance, a

method to fight against Sybil nodes (producing washes trades with a lot of transactions and high volume but low user quality, low impact on the ecosystem).

In this study, we presented the adaptive weighted PageRank algorithm, applied it to compute reputation scores of accounts within public blockchains. We validated the results with on-chain statistics and public awareness. In essence, reputation scores effectively capture the most important characteristics and social relations of Web3 (the Internet of Value) building on top of blockchains. Furthermore, onchain reputation can help enhancing DAO governance, mitigating Sybil attacks, ultimately establishing a Sybil-proof reputation system. In future studies, various user attributes can be explored and integrated into the PageRank’s data matrix. Moreover, as reputation ranking computation currently takes place in separate blockchain networks, there is potential for researching and implementing an integrated user reputation system across multiple blockchains.

REFERENCES

- [1] L. Page, S. Brin, R. Motwani, and T. Winograd, “The pagerank citation ranking: Bringing order to the web,” Technical Report 1999-66, Stanford InfoLab, 1999.
- [2] Fiocco, Melissa, “Banking for the Unbanked: The Promises, Pitfalls and Potentials of Mobile Banking”. <http://hdl.handle.net/10393/39526>
- [3] Athanasios N. Nikolakopoulos, John D. Garofalakis, “NCDawareRank: A novel ranking method that exploits the decomposable structure of the web,” Proceedings of the sixth ACM international conference on Web search and data mining (ACM WSDM ’13), Rome, Italy (2013). <https://doi.org/10.1145/2433396.2433415>
- [4] NEM: technical reference. https://docs.nem.io/pages/Whitepapers/NEM_techRef.pdf
- [5] Xiaoye Jiang, Lek-Heng Lim, Yuan Yao, Yinyu Ye, “Statistical ranking and combinatorial Hodge theory,” Math. Program, pp. 127-203, 2011. <https://doi.org/10.1007/s10107-010-0419-x>.
- [6] Thuat Do, Thao Nguyen and Hung Pham. “Delegated Proof of Reputation: a Novel Blockchain Consensus,” IECC’19 Proceedings of the 2019 International Electronics Communication Conference, pp. 90-98, Japan. <https://dl.acm.org/doi/10.1145/3343147.3343160>
- [7] Do, Huu-Dung, and Thuat Do. “PageRank and HodgeRank on Ethereum Transactions: A Measure for Social Credit,” International Journal of Software Innovation (IJSI) 11, no.1: 1-13. <http://doi.org/10.4018/IJSI.315737>
- [8] Dym, C.L., Wood, W.H. & Scott, M.J. Rank ordering engineering designs: pairwise comparison charts and Borda counts. Res Eng Design 13, 236–242 (2002). <https://doi.org/10.1007/s00163-002-0019-8>
- [9] Agryzkov, T., Oliver, J. L., Tortosa, L., & Vicent, J. F. (2012). An algorithm for ranking the nodes of an urban network based on the concept of PageRank vector. Applied Mathematics and Computation, 219(4), 2186-2193.
- [10] Agryzkov, T., Tortosa, L., Vicent, J. F., & Wilson, R. (2019). A centrality measure for urban networks based on the eigenvector centrality concept. Environment and Planning B: Urban Analytics and City Science, 46(4), 668-689.
- [11] Pillai, S. U., Suel, T., & Cha, S. (2005). The Perron-Frobenius theorem: some of its applications. IEEE Signal Processing Magazine, 22(2), 62-75.
- [12] Shanahan, J. G., & Dai, L. (2015, August). Large scale distributed data science using apache spark. In Proceedings of the 21th ACM SIGKDD international conference on knowledge discovery and data mining (pp. 2323-2324).

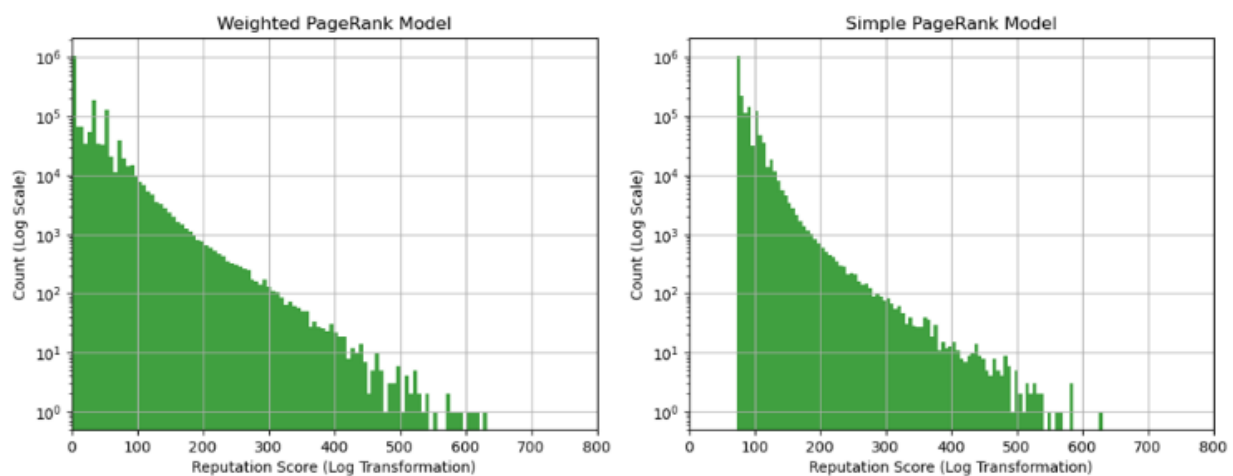


Fig. 1. Reputation scores distribution between our proposed adaptive weighted Pagerank (left) and the simple Pagerank (right), revealing similar shapes but different ranges.

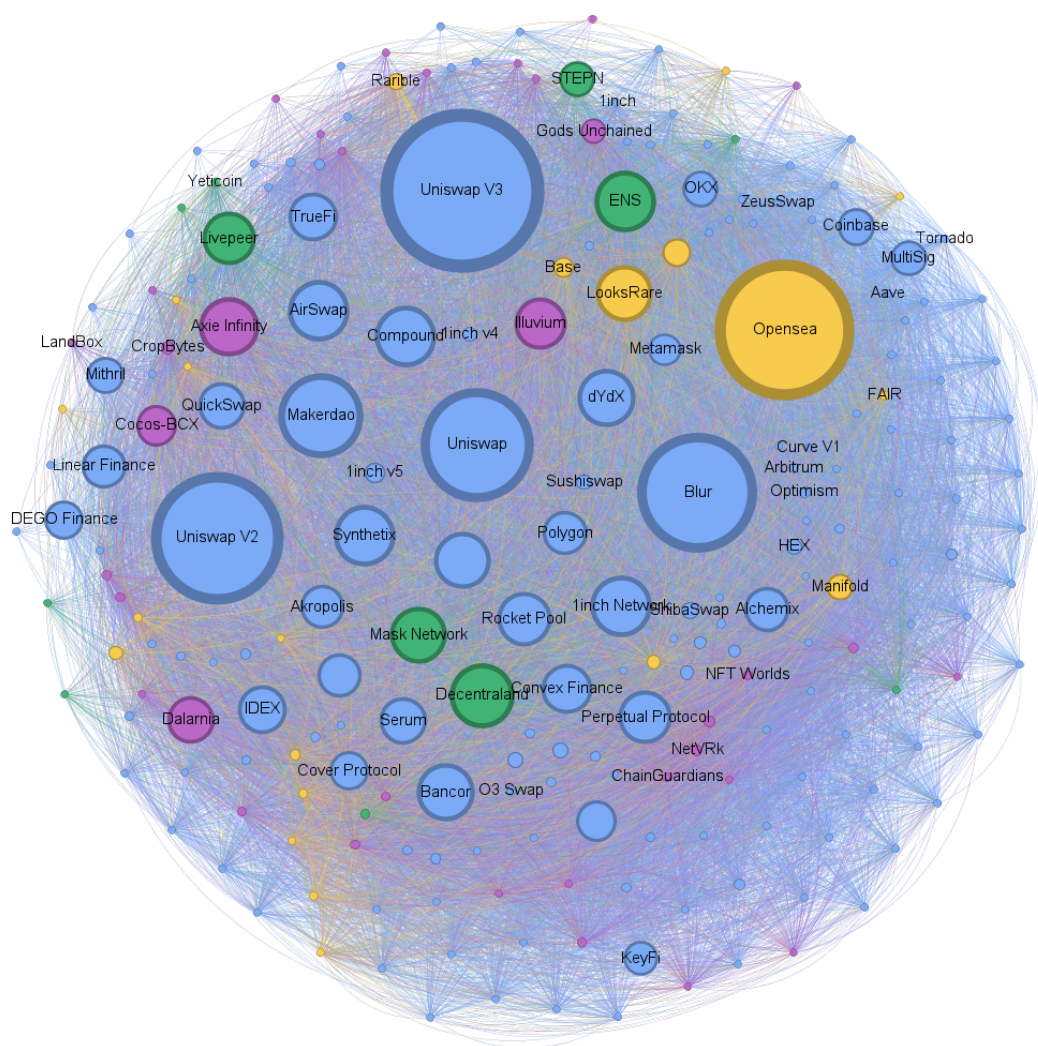


Fig. 2. Visualization of Dapps on the Ethereum network based on their users' accumulative reputation scores, with categories: DeFi (blue), NFT (yellow), social & metaverse (green), gamefi (violet).